Probability does not exist.

Bruno de Finetti

Bayesian inference is a predictive act. This book builds it from one idea, prediction, and shows that de Finetti's representation, Sanov's theorem and Bayes' rule are three views of a single calculation.

Fourteen chapters carry the argument from exchangeability, types and large deviations, through coding, shrinkage, maximum entropy, e-values and testing, to posterior asymptotics, scalable computation and deep learning, growth optimal investment, reinforcement learning, causal machine learning, medical decision making and federated learning.

Unlike treatments that present Bayes as a recipe for posteriors, this book treats every inference as a decision on one moderate deviation scale. Readers learn why a Bayesian test threshold grows with the sample size, how an e-value stays valid under continuous monitoring, why a universal portfolio is a posterior mean, and how a calibrated prior can change a cancer treatment decision.

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Elements of Bayes and Information Theory

Polson • Sokolov

Elements of Bayes and Information Theory

A Predictive Theory of Inference



Nick Polson and Vadim Sokolov